A060 v0.2.0
v1.0.7 Integration Candidate

M00–M04 analytic implementation and gate contract

SST-Workbench engineering delivery

17 October 2026

DEVELOPMENT / NOT CERTIFIED
No FREEZE, BASIC, FULL, CERTIFY or REVEAL performed.

1 Identification and lineage

This additive candidate was generated using the original frozen framework v1.0.7 generator with profile `multilibrary_gpu`. The 92 release files match the package manifest; the framework itself was not modified. Original development modules were migrated unchanged. One instance-local readiness guard was added to the generated launcher; its original source is retained for comparison.

Existing A060 v0.1.0, v0.2.0, v0.2.1 and v0.3.0 Drive material was not overwritten. A054 remains independent. The package must be extracted into an isolated review tree. Its v0.2.0 instance identity does not replace an existing frozen or development version.

2 Question and evidence boundary

Can the successful analytic development modules be integrated into a pinned v1.0.7 instance while preserving the native scientific gate meanings and explicitly retaining unexecuted requirements? The engineering answer is supported by the checks below. No physical claim follows.

3 Hypotheses

Physical H_0: verified local identities do not establish a self-confined, dynamically stable, free-standing three-dimensional vortex knot. A possible later alternative requires independently produced Euler dynamics beyond matched controls. This candidate cannot reject physical H_0.

Engineering predictions are narrower: Euler identities hold in the admissible region, tensor invariants satisfy their identities, projected forcing obeys Helmholtz closure, and affine spherical quadrature matches its local formula. A violation at a fixed control tolerance is an implementation-control failure.

4 Assumptions

The physical comparison model is unforced, incompressible, inviscid three-dimensional Euler flow. Modified forcing is a diagnostic comparison. Analytic fields, manufactured fields, numerical controls and independent physical providers are distinct evidence classes. The four source PDFs

supply equations, not independent discovery datasets. All scoring controls are dimensionless and contain no canonical particle constants.

5 Symbols and dimensions

With $A_{ij} = \partial_j u_i$ and $S = (A + A^T)/2$, the velocity gradient, strain and vorticity have units s^{-1}. The pressure invariant Q_A has units s^{-2}; R_A, R_s, V_s have units s^{-3}. The scale flux Φ_ℓ has units $\mathrm{m}^2\mathrm{s}^{-3}$. Code uses nondimensional coordinates and velocities. Restoring scales U_0, L_0 gives kinematic pressure $U_0^2 Q_L(\mathbf{x}/L_0)$ and momentum residual scale U_0^2/L_0.

6 Mathematical controls

The Landreman family is evaluated only for $s > 0$ and $F > 0$ [3]:

$$a = \sqrt{1+\epsilon}, \quad b = \sqrt{1-\epsilon}, \quad s = x^2/a^2 + y^2/b^2, \quad F = \sqrt{1-(1-s)^2-4z^2}, \tag{1}$$

$$\mathbf{B} = \left(\frac{2zx - (a/b)Fy}{s}, \frac{2zy + (b/a)Fx}{s}, 1-s \right), \qquad Q_L = (x^2+y^2+4z^2)/2. \tag{2}$$

The complete first two numerators are divided by s. Tensor pressure and shear controls follow Massey [2]. Projected vorticity-amplification controls follow the modified forcing in Chiarini [1]. Spherical flux controls follow Zhang and Yao [4]; their higher-mode remainder is retained for nonaffine fields.

Machine-readable science contract

Research question Do the preregistered M00-M04 analytic reference identities satisfy independent numeric implementations without physical SST target fitting?

Objective Qualify source provenance, a steady-Euler toroidal reference, pressure silence, VA/VT projection, and a spherical local-flux identity.

H_0 Analytic/numeric agreement alone does not demonstrate any stable particle, gravity or dynamic 3D Euler knot; scientific certification needs native and independent evidence.

H_1 A subsequent physical SST claim may be entertained only after independently produced 3D Euler fields pass native and dynamical qualification; not tested here.

Registered equations

E01: steady Euler reference

$$\nabla \cdot \mathbf{B} = 0, \ (\mathbf{B} \cdot \nabla)\mathbf{B} = -\nabla Q_L \tag{3}$$

Dimensional check: dimensionless normalized Landreman system

E02: pressure Poisson source

$$2Q_A = -\mathrm{tr}(A^2) = \omega^2/2 - S : S \tag{4}$$

Dimensional check: T^-2=T^-2

E03: shear wedge

$$m \leq 1 - f \leq 3m \tag{5}$$

Dimensional check: dimensionless

E04: VA VT projected source

$$\nabla \times \mathbf{g} = -\mathbb{P}_{\perp}\mathbf{VA} \tag{6}$$

Dimensional check: T^-2 after normalized reference

E05: local spherical scale flux

$$\Phi_{\ell} = \frac{\ell^2}{10}(3R_s + V_s/4) + \mathcal{O}_{\ell} \tag{7}$$

Dimensional check: L^2 T^-3

Registered code-to-mathematics steps

1. **S01** [G1; equations E01]: Validate SHA256 of original reference PDFs and explicitly log A045 registry mismatch
2. **S02** [G2; equations E01, E02]: Evaluate Landreman analytic field and central differences; compare momentum, divergence and pressure source
3. **S03** [G2; equations E02, E03]: Compute Q,R,Schur fraction f and m on analytic and seeded trace-free tensors
4. **S04** [G2; equations E04]: Use periodic FFT Poisson projection, compare curl force with -PVA, integrate correction
5. **S05** [G2; equations E05]: Integrate exact affine velocity increments on sphere and compare with gradient-flux identity

7 Implemented engineering algorithm

Run original six regression controls and additional source, schema, gate-map, guard and boundary tests. Compare centered differences against an independently coded analytic Jacobian on 81 volume points at $\epsilon = 0, 0.2, 0.5$, with spacings $10^{-3}, 5\times10^{-4}, 2.5\times10^{-4}$. Test 2500 seeded trace-free tensors, 64 rotations, rank perturbations over 13 amplitudes, odd Fourier grids 17/25/33 and spherical quadrature over three diameters and three angular resolutions.

For a manufactured solenoidal cubic field $\mathbf{u} = (y + \beta y^2 z, z, x)$, symmetric increments retain a nonzero higher-mode flux at the origin. The measured remainder is reported separately from the zero local gradient prediction. It is not a turbulence simulation. The C++17 double module independently computes tensor-only invariants on 29 cases using the shared framework build machinery.

8 Sources and identity status

All four original PDF hashes match their 13 October source lock. The A060 family folder is verified. An existing v0.3.0 README proposes A045 as Two-Harmonic Phase-Chirality Floquet and A046 as Relative Vorticity Triadic Geometry. This lead has not been reconciled against an authoritative registry snapshot: identity authority remains `UNRESOLVED`. No registry was edited. No A054 providers or verdicts were imported.

9 Native gates and mapping

M00 maps to G0/G1. M01, M02, M03 and M04 all map to G2. Preserve the native chain G0 through G9; do not reinterpret analytic controls as G3 discovery.

Gate	Current state	Required closure
G0	UNRESOLVED	Completed campaign readiness and native freeze; matching implementation lock.
G1	UNRESOLVED	Authoritative registry/identity reconciliation and producer-qualified sources.
G2	UNRESOLVED	Frozen production parameters and actual eligible M01–M04 execution.
G3	UNRESOLVED	Independent blind discovery provider and statistic.
G4	UNRESOLVED	Independent held-out confirmation provider.
G5	UNRESOLVED	Complete preregistered native parity; tensor preflight is partial.
G6	UNRESOLVED	Actual enabled GPU screening; no FP64 promotion.
G7	UNRESOLVED	Independent provenance-group replication.
G8	UNRESOLVED	Qualified 3D Euler dynamics and confinement/stability convergence.
G9	UNRESOLVED	Eligible sealed blind results and both verified nonced commitments.

An actual blind framework run records G9 as DEFERRED and blocked dependencies using native statuses; neither means scientific PASS. No official gate ledger was fabricated for this unfrozen candidate.

10 Numerical tolerances and open definitions

Development thresholds remain reference controls: divergence 3×10^{-7}, momentum 2×10^{-7}, pressure-Poisson 3×10^{-6}, wedge 10^{-10}, projector 10^{-8}, integrated correction 10^{-9} and affine quadrature 10^{-12}. The added engineering checks use rotation tolerance 10^{-10}, analytic residual 10^{-11} and refinement ratio greater than 3.5. These choices have not been promoted to a frozen production campaign. Discovery statistics, full production grids, confirmatory nulls, full backend parity and physical criteria remain UNRESOLVED.

11 Backend authority

Python/NumPy FP64 is REFERENCE. Actual C++ double tensor execution is recorded with compiler fingerprint, ABI, scope and parity error. It does not certify the complete M01–M04

pipeline or dynamics. SYCL FP32 and DD32 remain SCREENING ONLY and were not run here. Missing native execution never falls back to a Python certification PASS.

12 Blindness and freeze guard

The existing development controls were already observed. Their new checks are engineering diagnostics, not newly blinded evidence. No scientific target identities are scored. Private generator secrets are excluded from the public source archive. New prospective key material and reveal payload are required before a later frozen campaign.

The instance-local readiness guard blocks all run modes except SELFTEST until required items have auditable resolutions. The science contract includes the implementation-lock digest; the guard checks its code files. No freeze commitment was generated. A complete schema alone cannot satisfy scientific readiness.

13 Statistics and edge cases

Analytic tolerances are numerical error limits, not significance levels. Gaussian tensors and rotations are controls rather than independent providers. Zero tensor normalized invariants are undefined; zero-vorticity VA is masked; invalid Landreman points are rejected. Fourier zero mean is explicit. Refinement is demonstrated for the stated samples and spacings only; no general numerical error bound is claimed.

14 Current engineering results

Engineering check	Observed result
Framework release bytes	92/92 MATCH
Framework SELFTEST	56/56 PASS
Instance regression	18/18 PASS
Paper hashes	4/4 MATCH
Volume refinement samples	81
Finite-difference error ratios	3.999990; 3.999998
Analytic Euler residual	3.553e-15
Rotation invariant error	1.066e-14
Rank-one fitted exponent	0.500000
Rank-two fitted exponent	0.333333
Spherical quadrature error	3.242e-15
Nonaffine higher-mode remainder	-0.002058
Native tensor scaled parity	8.882e-16
Native tensor scope	29 cases; Q, pressure, Rs, Vs, fluxfactor

No row above closes an official scientific gate. Detailed JSON evidence, source locks and migration hashes accompany this report.

15 Interpretation and limitations

The requested instance structure, conservative gate mapping and executable engineering checks are implemented. Scientific status remains DEVELOPMENT / NOT CERTIFIED. Registry authority,

prospective blindness, independent providers, complete native parity, GPU results and physical stability remain UNRESOLVED.

Analogy: checking that a ruler and a stopwatch work does not yet show that a spinning loop can hold itself together.

16 Execution order and deliverables

Run SELFTEST, then the separate instance tests and precheck. Only after readiness is resolved may the sequence continue: FREEZE, BASIC, FULL, CERTIFY, REVEAL_IF_ALLOWED. Preserve each sealed blind mode snapshot before a rerun. Reveal consumes an existing ledger without rerunning science; inspect decision contents and integrity, not exit status alone.

The public integration ZIP contains the exact framework source, candidate instance, tests, PDF references, contracts and engineering evidence. Extract it into an isolated review directory. Existing Drive releases and their historical results were left unchanged. The report PDF is derived from this TeX source and is not a native frozen falsifier result.

References

[1] A. Chiarini (2026), "Is Vorticity Amplification Essential for Anomalous Dissipation and Intermittency in Turbulence?", arXiv:2609.39826v1, https://arxiv.org/abs/2609.39826.

[2] J. M. O. Massey (2026), "Near-cancellation of enstrophy and strain requires shear or a balanced vortex", arXiv:2609.32543v1, https://arxiv.org/abs/2609.32543.

[3] M. Landreman (2026), "Analytic toroidal 3D MHD equilibria and steady Euler flows with invariant surfaces", arXiv:2609.26742v2, https://arxiv.org/abs/2609.26742.

[4] W. Zhang and H. Yao (2026), "Geometric Mechanics of Real-Space Energy Transfer in Turbulence", arXiv:2610.11029v1, https://arxiv.org/abs/2610.11029.